Exam E3.17 2022-23 Answers

Q1

Q1)

a)

i) For the case $a>1$: at t_0+1 the frame has been transmitted but the sending station needs to wait for it reception. This happens at t_0+a. At this point a free token is released.
For the case $a<1$: at t_0+a the frame is being received but the complete frame has not been transmitted. The frame is transmitted at t_0+1. At this point a free token is released.

ii) use $a = \frac{Rd}{LV}$ $\quad V = 0.66 \times 2.99 \times 10^8$ m/s

R	L	d	→	a	$u = 1/1+a$
			calculate		
1	100	1		0.05	0.95
1	100	10		0.5	0.67
10	1000	10		0.5	0.67
100	1000	50		25	0.04

discuss the impact of R, L, d on a and $u = 1/1+a$

Q1 b)

$K \sim 2a+1$ if $W > 2a+1$

$K \sim W$ if $W < 2a+1$

Time / delay sequence diagram can be used to show the bounds / estimations.

Q1

c)

i) - IP address
- subnet mask
- address to nearby router

ii) DHCP : Dynamic Host configuration Protocol

To allow host to dynamically obtain its IP address from the server when it joins the network.

DHCP provides
- IP address
- address to first hop router
- Name and IP address of DNS server
- Network mask

Introduce - the DNS server and function
- resolve addresses
- email services support

Answers

Q2

Q2.1

i) The following or similar table can be used to explain differences:

Deterministic sh path	Statistical sh path
link cost = l_i	link cost = (μ_i, σ_i^2)
Path cost = $\sum l_i$	Path cost = $f(\mu_p, \sigma_p^2)$
Minimize $\sum l_i$	Minimize $\mu_p + \phi(\sigma_p^2)$

Example 1: Vehicular transport network under unexpected traffic conditions

Example 2: Temperature aware routing in microchip.

ii) ICMP: Internet Control Message Protocol
- communicate network-layer information e.g. error reporting, Example: destination network unreachable.

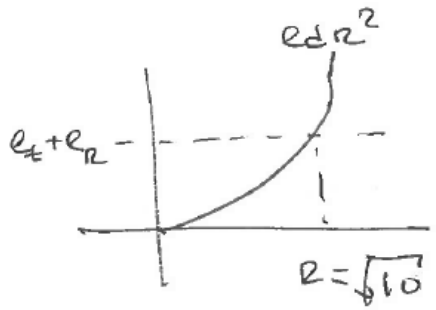
ICMP messages are carried inside IP datagrams (as payload). The extracted is similar as TCP or UDP content to upper layers.

Answers

Q2.2)

i) $E_t + E_R = e_t + e_d R^2 + e_R$

$= (2 \times 5 \times 10^{-10} + 100 \times 10^{-12} \times R^2)$



$R^2 = 10$

$R = \sqrt{10}$

ii) At the distance shown in fig 2.1
All path have the same cost.
Discussion of other possible features that will be worth consideration:
- Reliability
- mobility pattern of nodes
- number of nodes in the path.
-

iii) the added node should be placed in the middle of the direct line between ① and ⑥

Discussion

Answers

Q3

Q3.1

i) Discuss the following principle and related network mechanisms:

- service level agreement
- Policing mechanism
- call admission
- efficient use of bandwidth

Q3.2

i) Network cost $= \frac{x(1)}{c(1)+x(1)} + \frac{x(2)}{c(2)+x(2)} + \frac{a}{x(1)+x(2)}$

ii) $\frac{c(1)}{(c(1)-x(1))^2} = \frac{a}{(x(1)+x(2))^2} = \frac{a}{x^2}$ link 1

$\frac{c(2)}{(c(2)-x(2))^2} = \frac{a}{(x(1)+x(2))^2} = \frac{a}{x^2}$ link 2

$\frac{a}{x}$ influence how much flow can be kept out of the network

iii) $c(1) > c(2)$ link 1 will start carrying traffic

$x(2) = \phi$ $\frac{c(1)}{(c(1)-x(1))^2} \leq \frac{a}{x(1)^2}$

$x(1) \leq \frac{\sqrt{a}\, c_1}{\sqrt{c_1}+\sqrt{a}}$

After this point link 2 will start carry traffic as $x(2) \neq \phi$.

Q4

Q 4.1

i) A proportional fair algorithm attempts to allocate flows in proportion to their request x_i.
The solution is given by $\max \sum_i x_i$
equivalent to $\max \sum \ln x_i$

ii) $$\lambda + \lambda_0 + \frac{\lambda_0}{3} = 1 \longrightarrow \lambda = 1 - \frac{4}{3}\lambda_0$$

$$\ln\lambda + 2\ln\lambda_0 + \ln\frac{\lambda_0}{3}$$

$$\frac{d}{d\lambda_0}\left(\ln\left(1-\frac{4}{3}\lambda_0\right) + 2\ln\lambda_0 + \ln\frac{\lambda_0}{3}\right) = 0$$

$$\frac{-4/3}{\left(1-\frac{4}{3}\lambda_0\right)} + \frac{2}{\lambda_0} + \frac{1}{\lambda_0} = 0$$

$$4\lambda_0 = 3\times 3\left(1-\frac{4}{3}\lambda_0\right)$$

$$4\lambda_0 = 9 - 12\lambda_0$$

$$\boxed{\lambda_0 = \frac{9}{16}}$$

and

$$\lambda = 1 - \frac{4}{3}\frac{9}{16} = \frac{3}{4} = \frac{4}{16}$$

$$\boxed{\lambda = \frac{4}{16}}$$

Q4.2

a) RTS / CTS

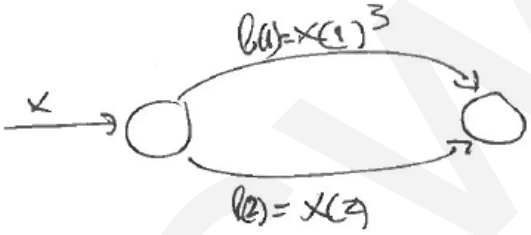
i) This is a prefixed reservation scheme to avoid collisions even in the presence of hidden terminals.

- collision avoidance RTS/CTS sequence diagram

ii) RTS/CTS . introduce delays and consumes channel resources

The RTS/CTS sequence is useful only when is required to transmit long data frames

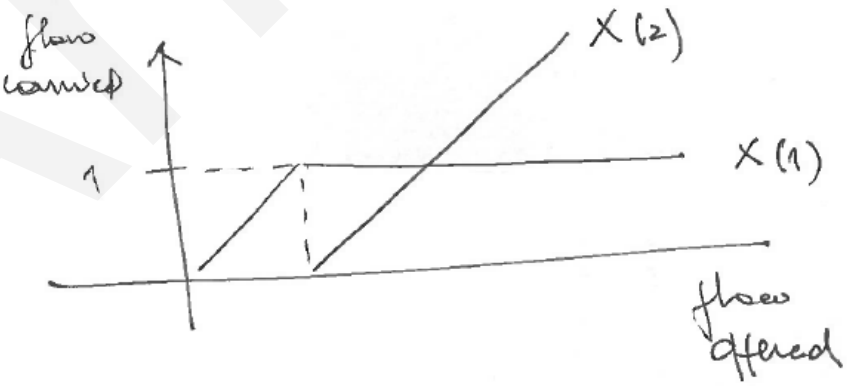
b)



$X = x(1) + x(2)$

i) optimality condition $3x(1)^2 = 1$

ii)



Answers